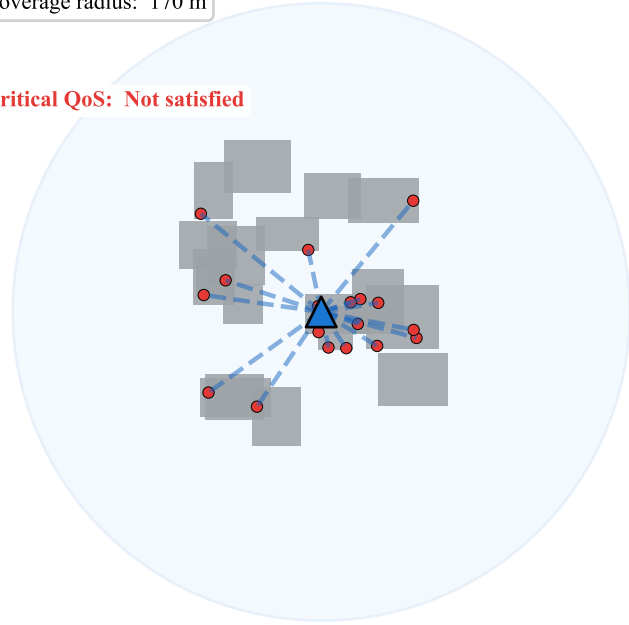


Altitude → Coverage Radius → Communication Quality

2D-AUTO

Altitude: 100 m
Coverage radius: 170 m

Critical QoS: Not satisfied

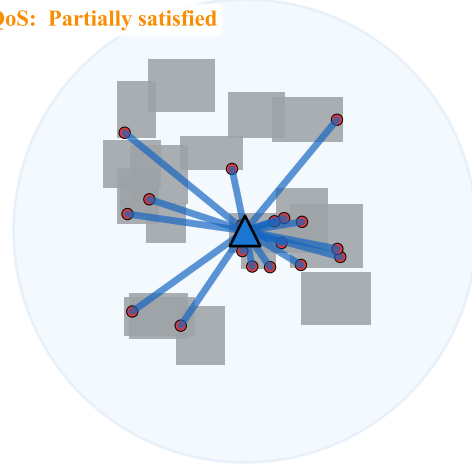


Large footprint
Low link quality

3D-GNN

Altitude: 75 m
Coverage radius: 127 m

Critical QoS: Partially satisfied

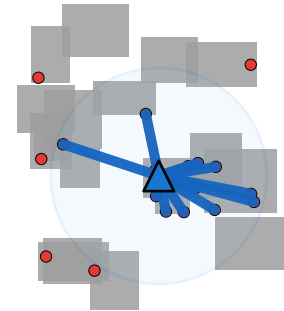


Moderate footprint
Limited adaptation

ATOM-3D-VoI (Ours)

Altitude: 35 m
Coverage radius: 59 m

Critical QoS: Satisfied



Focused footprint
High QoS for critical nodes

● High priority ● Medium priority ● Low priority ▲ UAV — Critical link (thicker = higher quality) Coverage footprint